

PROGRAMMING FUNDAMENTALS

Spring 2024

25th April

Ex-1

- Write a function that rolls a pair of dice until the sum of the numbers rolled is a specific number
- We also want to know the number of times the dice are rolled to get the desired sum
- Do not forget to include the following
- `#include <cstdlib>`
- `#include <ctime>`

Function

```
int rollDice(int num)
{
    int die1;
    int die2;
    int sum;
    int rollCount = 0;

    srand(time(0));

    do
    {
        die1 = rand() % 6 + 1;
        die2 = rand() % 6 + 1;
        sum = die1 + die2;
        rollCount++;
    }
    while (sum != num);

    return rollCount;
}
```

Main function

```
int main()
{
    cout << "The number of times the dice are rolled to "
          << "get the sum 10 = " << rollDice(10) << endl;
    cout << "The number of times the dice are rolled to "
          << "get the sum 6 = " << rollDice(6) << endl;

    return 0;
}
```

Sample Run

The number of times the dice are rolled to get the sum 10 = 11

The number of times the dice are rolled to get the sum 6 = 7

Ex-2 - Function

```
double larger(double x, double y)
{
    if (x >= y)
        return x;
    else
        return y;
}
```

Main Function

```
double larger(double x, double y);

int main()
{
    double num; //variable to hold the current number
    double max; //variable to hold the largest number
    int count;  //loop control variable

    cout << "Enter 10 numbers." << endl;
    cin >> num;
    max = num;

    for (count = 1; count < 10; count++)
    {
        cin >> num;
        max = larger(max, num);
    }

    cout << "The largest number is " << max
         << endl;

    return 0;
} //end main
```

Output

- ???????

Output

```
Enter 10 numbers.
```

```
20 36 71 89 11 65 55 90 44 65
```

```
The largest number is 90
```